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SEMICONDUCTOR DEVICE AND METHOD OF MANUFACTURING SEMICONDUCTOR DEVICE

Abstract

A semiconductor device may include a first semiconductor structure including a peripheral circuit and an interconnection structure electrically connected to the peripheral circuit, a second semiconductor structure, a bonding layer positioned between the first and second semiconductor structures, a bonding pad electrically connected to the interconnection structure, the bonding pad extending into the bonding layer, and a contact plug extending into the bonding layer through the second semiconductor structure, wherein the contact plug is electrically connected to the bonding pad.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0032127 filed on Mar. 6, 2024, which is incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] Embodiments of the present disclosure relate to an electronic device, and more particularly, to a semiconductor device and a method of manufacturing the semiconductor device.

2. Related Art

[0003] An integration degree of a semiconductor device is mainly determined by a region occupied by a unit memory cell. Recently, as improvements in the integration degree of a semiconductor device in which a memory cell is formed as a single layer on a substrate reach a limit, a three-dimensional semiconductor device in which memory cells are stacked on a substrate is being proposed. In addition, various structures and manufacturing methods are being developed in order to improve operation reliability of the semiconductor device.

SUMMARY

[0004] According to an embodiment of the present disclosure, a semiconductor device may include a first semiconductor structure including a peripheral circuit and an interconnection structure electrically connected to the peripheral circuit, a second semiconductor structure, a bonding layer positioned between the first and second semiconductor structures, a bonding pad electrically connected to the interconnection structure, the bonding pad extending into the bonding layer, and a contact plug extending into the bonding layer through the second semiconductor structure, wherein the contact plug is electrically connected to the bonding pad.

[0005] According to an embodiment of the present disclosure, a semiconductor device may include a first bonding layer, a first bonding pad passing through the first bonding layer, a second bonding layer forming a bonding interface with the first bonding layer, a first gate structure on the second bonding layer, and a first contact plug extending through the first gate structure and the second bonding layer and electrically connected to the first bonding pad, and the first contact plug may include a metal layer, and a barrier layer surrounding a sidewall of the metal layer and extending between the metal layer and the first bonding pad.

[0006] According to an embodiment of the present disclosure, a method of manufacturing a semiconductor device may include forming a first wafer including a first bonding layer and a first bonding pad, forming a second wafer including a second bonding layer and a first sacrificial plug, bonding the first bonding layer and the second bonding layer so that the first bonding pad and the first sacrificial plug are connected, forming a first opening exposing the first bonding pad by removing the first sacrificial plug, and forming a first contact plug electrically connected to the first bonding pad, in the first opening.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIGS. 1A and 1B are diagrams illustrating a structure of a semiconductor device according

to an embodiment of the present disclosure.

[0008] FIGS. 2A and 2B are cross-sectional views illustrating a structure of a semiconductor device according to an embodiment of the present disclosure.

[0009] FIG. 3 is a diagram illustrating a structure of a semiconductor device according to an embodiment of the present disclosure.

[0010] FIGS. 4A to 4C are diagrams illustrating a structure of a semiconductor device according to an embodiment of the present disclosure.

[0011] FIGS. 5A to 5D are diagrams illustrating a method of manufacturing a semiconductor device according to an embodiment of the present disclosure.

[0012] FIGS. 6A to 6C are diagrams illustrating a method of manufacturing a semiconductor device according to an embodiment of the present disclosure.

[0013] FIGS. 7A to 7C are diagrams illustrating a method of manufacturing a semiconductor device according to an embodiment of the present disclosure.

[0014] FIGS. 8A to 8G are diagrams illustrating a method of manufacturing a semiconductor device according to an embodiment of the present disclosure.

[0015] FIG. 9 is a configuration diagram of a semiconductor device according to an embodiment of the present invention disclosure.

[0016] FIG. 10 is a configuration diagram of a semiconductor device according to an embodiment of the present invention disclosure.

DETAILED DESCRIPTION

[0017] Embodiments of the present disclosure provide a semiconductor device and a method of manufacturing the semiconductor device having a stable structure and improved performance characteristics.

[0018] The integration degree of the semiconductor device is improved by stacking memory cells in a three-dimensional structure. In addition, the semiconductor device with the three-dimensional structure has a stable structure, and exhibits improved reliability and performance.

[0019] Hereinafter, embodiments according to the technical concepts of the present disclosure are described with reference to the accompanying drawings.

[0020] FIGS. 1A and 1B are diagrams illustrating a structure of a semiconductor device according to an embodiment of the present disclosure.

[0021] Referring to FIG. 1A, the semiconductor device may include a first semiconductor structure SS1, a second semiconductor structure SS2, and a bonding structure BS disposed between the first and second semiconductor structures SS1, and SS2. The semiconductor device may further comprise at least one contact plug 16, e.g., two or more contact plugs 16. For example, the first semiconductor structure SS1 and the second semiconductor structure SS2 may include a peripheral circuit and/or a memory cell array. In an embodiment, the first semiconductor structure SS1 may include the peripheral circuit. The first semiconductor structure SS1 may further include a first interconnection structure IC1 electrically connected to the peripheral circuit, an interlayer insulating layer, and the like. In an embodiment, the second semiconductor structure SS2 may include the memory cell array. The second semiconductor structure SS2 may include a source structure, a gate structure, a channel structure, a second interconnection structure, an interlayer insulating layer, and the like. The memory cell array may include memory cells stacked along a channel structure. In an embodiment, the first semiconductor structure SS1 may include the memory cell array, and the second semiconductor structure SS2 may include the peripheral circuit.

[0022] The bonding structure BS electrically connects the first semiconductor structure SS1 and the second semiconductor structure SS2 by bonding the first semiconductor structure SS1 and the second semiconductor structure SS2. The bonding structure BS may include a bonding layer B and first and second bonding insulating layers 13 and 14. The bonding layer B may be disposed between the first and second bonding insulating layers 13 and 14. The bonding layer B may include a first bonding layer 11 and a second bonding layer 12. At least one bonding pad 15, e.g., two or

more bonding pads **15** may be disposed within the bonding structure **BS**.

[0023] The first and second bonding layers **11** and **12** may form a bonding interface **IF**. The first bonding layer **11** may be positioned over the first semiconductor structure **SS1**. The first bonding layer **11** may be positioned on the first bonding insulating layer **13**. The second bonding layer **12** may be positioned between the first bonding layer **11** and the second bonding insulating layer **14**. The first bonding insulating layer **13** may be positioned between the first semiconductor structure **SS1** and the first bonding layer **11**. The second bonding insulating layer **14** may be positioned between the second semiconductor structure **SS2** and the second bonding layer **12**. The bonding interface **IF** may be positioned at a portion where the first and second bonding layers **11** and **12** are in contact with each other. The first and second bonding layers **11** and **12** may be chemically bonded at the bonding interface **IF**.

[0024] In an embodiment, the first and second bonding layers **11** and **12** may include tetraethylorthosilicate (TEOS), silicon carbon nitride (SiCN), or the like. When bonding the first and second bonding layers **11** and **12** formed of a SiCN layer, Si—O—Si bonding may be generated on a surface of the first and second bonding layers **11** and **12**, and the bonding interface **IF** may be changed to an oxide such as tetraethylorthosilicate (TEOS) by removing carbon and nitrogen from SiCN.

[0025] The first bonding insulating layer **13** may be positioned between the first semiconductor structure **SS1** and the first bonding layer **11** for protecting the first bonding layer **11** in a manufacturing process. The first bonding insulating layer **13** may be a single layer. The first bonding insulating layer **13** may be multiple layers. The second bonding insulating layer **14** may be positioned between the second semiconductor structure **SS2** and the second bonding layer **12** for protecting the second bonding layer **12** in the manufacturing process. The second bonding insulating layer **14** may be a single layer. The second bonding insulating layer **14** may be multiple layers. The first and second bonding insulating layers **13** and **14** may include the same or different insulating material such as, for example, an oxide, a nitride, or TEOS.

[0026] The bonding pad **15** may extend into the bonding layer **B** and may pass through the first bonding layer **11**. In an embodiment, the bonding pad **15** may extend through the first bonding layer **11** and the first bonding insulating layer **13** to electrically connect to the first interconnection structure **IC1**. The bonding pad **15** may be electrically connected to the peripheral circuit of the first semiconductor structure **SS1** through the first interconnection structure **IC1**.

[0027] The contact plug **16** may extend into the bonding layer **B** through the second semiconductor structure **SS2**, and may pass through the second bonding layer **12**. In an embodiment, the contact plug **16** may extend through the gate structure, the source structure, the interlayer insulating layer, and the like of the second semiconductor structure **SS2**, and may pass through the second bonding insulating layer **14**. The contact plug **16** may be electrically connected to the bonding pad **15** at the bonding interface **IF**.

[0028] Referring to FIG. **1B**, the contact plug **16** may include a barrier layer **16A** and a metal layer **16B** in the barrier layer **16A**. The barrier layer **16A** may surround a sidewall of the metal layer **16B** and may extend between the metal layer **16B** and the bonding pad **15**. The barrier layer **16A** may be in contact with the bonding pad **15**. The barrier layer **16A** may have a substantially uniform thickness along its entire extent.

[0029] According to the structure described above, the bonding structure **BS** may include the bonding pad **15** corresponding to the first semiconductor structure **SS1** and may not include a bonding pad corresponding to the second semiconductor structure **SS2**. Instead, the contact plug **16** passing through the second semiconductor structure **SS2** may be directly connected to the bonding pad **15**. The contact plug **16** may pass through the second semiconductor structure **SS2**, the second bonding insulating layer **14**, and the second bonding layer **12**, and may be directly connected to the bonding pad **15**.

[0030] FIGS. **2A** and **2B** are cross-sectional views illustrating a structure of a semiconductor device

according to an embodiment of the present disclosure. FIGS. 2A and 2B are enlarged views of a region A of FIG. 1A. Hereinafter, a description that overlaps the content described above may be omitted.

[0031] Referring to FIG. 2A, the semiconductor device may include a first bonding layer **11**, a second bonding layer **12**, a first bonding insulating layer **13**, a second bonding insulating layer **14**, a bonding pad **15**, and a contact plug **16**. The contact plug **16** may pass through the second bonding insulating layer **14** and the second bonding layer **12** and may extend into the first bonding layer **11**.

[0032] A lower surface of the contact plug **16** may be in contact with the bonding pad **15**. A contact surface of the contact plug **16** and the bonding pad **15** may be flat or may include a curved surface. In an embodiment, an upper surface of the bonding pad **15** may be positioned lower than an upper surface of the first bonding layer **11** and may include a curved surface. This structure may be due to dishing induced in a planarization process. A step may exist between the upper surface of the bonding pad **15** and the upper surface of the first bonding layer **11** due to dishing, and the step may be filled with the contact plug **16**.

[0033] The contact plug **16** may include a through portion **16C** and an expansion portion **16D**. The through portion **16C** may pass through the second bonding layer **12** and the second bonding insulating layer **14**, and may extend into the second semiconductor structure SS2. The through portion **16C** may have a cross-section of a taper shape or a bowing shape. The expansion portion **16D** may fill the step between the bonding pad **15** and the first bonding layer **11**. The expansion portion **16D** may extend into the first bonding layer **11** and may have a width greater than that of the through portion **16C**. The expansion portion **16D** may have a flat upper surface at the bonding interface IF, and a lower surface that is in contact with the bonding pad **15** may include a curved surface.

[0034] Referring to FIG. 2B, the semiconductor device may include a first bonding layer **11**, a second bonding layer **12**, a first bonding insulating layer **13**, a second bonding insulating layer **14**, a bonding pad **15**, and a contact plug **16**. The contact plug **16** may pass through the second bonding insulating layer **14** and the second bonding layer **12**, and may extend into the first bonding layer **11**.

[0035] A foreign substance **17** may exist between the bonding pad **15** and the second bonding layer **12**. The foreign substance **17** may be a by-product caused in the manufacturing process. In an embodiment, the foreign substance **17** may be an etching by-product or a bonding by-product, or may be a polymer-based material. The foreign substance **17** may not exist between the bonding pad **15** and the contact plug **16**.

[0036] According to the structure described above, the contact plug **16** may be expanded into the first bonding layer **11**. A lower surface of the contact plug **16** may include a curved surface and may protrude into the bonding pad **15**. Therefore, a contact area of the bonding pad **15** and the contact plug **16** may be increased, and the foreign substance **17** may be removed between the bonding pad **15** and the contact plug **16** to improve a bonding defect.

[0037] FIG. 3 is a diagram illustrating a structure of a semiconductor device according to an embodiment of the present disclosure. Hereinafter, a description that overlaps the content described above may be omitted.

[0038] Referring to FIG. 3, the semiconductor device may include a first semiconductor structure SS1, a second semiconductor structure SS2, a bonding structure BS, and contact plugs **36A** and **36B**. The bonding structure BS may be positioned between the first semiconductor structure SS1 and the second semiconductor structure SS2.

[0039] The first semiconductor structure SS1 may include a substrate **30**, peripheral circuits PC1 and PC2, a first interlayer insulating layer IL1, and a first interconnection structure IC1. The first peripheral circuit PC1 may be a page buffer, and the second peripheral circuit PC2 may be a row decoder. The first interconnection structure IC1 may be positioned in the first interlayer insulating layer IL1. The first interconnection structure IC1 may be electrically connected to the peripheral circuits PC1 and PC2 and may include a via and/or a line.

[0040] The second semiconductor structure SS2 may include a source structure S, a gate structure GST, a channel structure CH, a slit structure SLS, a second interlayer insulating layer IL2, and a second interconnection structure IC2. A gate structure GST may be positioned on the source structure S, and the gate structure GST may include gate lines 37 and insulating layers 38 that are alternately stacked. In an embodiment, the gate lines 37 may be a source select line, a word line, and or a drain select line. Each of the channel structures CH may extend through the gate structure GST and may be connected to the source structure S. The slit structure SLS may extend through the gate structure GST and may include an insulating material, a semiconductor material, or a conductive material.

[0041] The second interconnection structure IC2 may be positioned in the second interlayer insulating layer IL2. The second interconnection structure IC2 may be connected to the channel structure CH, the slit structure SLS, the gate lines 37, and the like, and may include a via and/or a line. In an embodiment, the second interconnection structure IC2 may include one or more bit lines connected to the channel structures CH.

[0042] The bonding structure BS may include a first bonding layer 31, a second bonding layer 32, a first bonding insulating layer 33, a second bonding insulating layer 34, and bonding pads 35A and 35B. The contact plugs 36A and 36B may pass through the second semiconductor structure SS2, and may be directly connected to the bonding pads 35A and 35B. The peripheral circuits PC1 and PC2 and the memory cell array CA may be electrically connected through the contact plugs 36A and 36B and the bonding pads 35A and 35B.

[0043] In an embodiment, the first contact plug 36A may be connected to the bit line through the second interconnection structure IC2, and may extend through the gate structure GST and the source structure S. The first contact plug 36A may pass through the second bonding insulating layer 34 and the second bonding layer 32 to electrically connect to the first bonding pad 35A. In this case, the bit line of the memory cell array CA and the page buffer of the first peripheral circuit PC1 may be connected through the first contact plug 36A and the first bonding pad 35A.

[0044] In an embodiment, the second contact plug 36B may be connected to the gate line 37 through the second interconnection structure IC2, and may extend through the gate structure GST. The second contact plug 36B may pass through the second bonding insulating layer 34 and the second bonding layer 32 to electrically connect to the second bonding pad 35B. In this case, the gate line 37 of the memory cell array CA and the row decoder of the second peripheral circuit PC2 may be connected through the second contact plug 36B and the second bonding pad 35B.

[0045] According to the structure described above, the contact plugs 36A and 36B passing through the second semiconductor structure SS2 may extend into the bonding structure BS, and may be directly connected to the bonding pads 35A and 35B. Therefore, the bonding structure BS may be simplified.

[0046] FIGS. 4A to 4C are diagrams illustrating a structure of a semiconductor device according to an embodiment of the present disclosure. Hereinafter, a description that overlaps the content described above may be omitted.

[0047] Referring to FIG. 4A, the semiconductor device may include first, second, and third semiconductor structures SS1, SS2, and SS3, first and second bonding structures BS1, and BS2, and contact plugs 36A, 36B1, 36B2, 46A, and 46B. The first bonding structure BS1 may be positioned between the first and second semiconductor structures SS1 and SS2. The second bonding structure BS2 may be positioned between the second and third semiconductor structures SS2 and SS3.

[0048] The first semiconductor structure SS1 may include a substrate 30, peripheral circuits PC1 and PC2, a first interlayer insulating layer IL1, and a first interconnection structure IC1. The first interlayer insulating layer IL1 may be formed on the substrate 30, and the first interconnection structure IC1 may be formed within the first interlayer insulating layer IL1. The second semiconductor structure SS2 may include a source structure S, a first gate structure GST1, a

channel structure CH, a slit structure SLS, a second interlayer insulating layer IL2, and a second interconnection structure IC2. The first gate structure GST1 may include gate lines 37 and insulating layers 38 that are alternately stacked, and may include a first step structure.

[0049] The third semiconductor structure SS3 may include a source structure S, a second gate structure GST2, a channel structure CH, a slit structure SLS, a third interlayer insulating layer IL3, and a third interconnection structure IC3. A second gate structure GST2 may be positioned on the source structure S, and the second gate structure GST2 may include gate lines 47 and insulating layers 48 that are alternately stacked. In an embodiment, the gate line 47 may be a source select line, a word line, or a drain select line. The second gate structure GST2 may include a second step structure. The first step structure and the second step structure may be arranged in a staggered manner so as not to overlap.

[0050] The third interconnection structure IC3 may be positioned in the third interlayer insulating layer IL3. The third interconnection structure IC3 may be connected to the channel structure CH, the slit structure SLS, the gate lines 47, and the like, and may include a via and/or a line. In an embodiment, the third interconnection structure IC3 may include a bit line connected to the channel structure CH.

[0051] The first bonding structure BS1 may include first and second bonding layers 31 and 32, first and second bonding insulating layers 33 and 34, and bonding pads 35A, 35B1, and 35B2. The second bonding structure BS2 may include third and fourth bonding layers 41 and 42, third and fourth bonding insulating layers 43 and 44, and bonding pads 45A and 45B.

[0052] The contact plugs 46A and 46B may extend through the third semiconductor structure SS3 and may be directly connected to the bonding pads 45A and 45B. A first memory cell array CA1 and a second memory cell array CA2 may be electrically connected through the contact plugs 46A and 46B and the bonding pads 45A and 45B.

[0053] In an embodiment, the third contact plug 46A may be electrically connected to the bit line, and may extend through the second gate structure GST2 and the source structure S. The third contact plug 46A may pass through the fourth bonding insulating layer 44 and the fourth bonding layer 42 to electrically connect to the third bonding pad 45A. The third bonding pad 45A may be connected to the first contact plug 36A through the second interconnection structure IC2. In this case, the third contact plug 46A, the third bonding pad 45A, the first contact plug 36A, and the first bonding pad 35A may be electrically connected. Therefore, a bit line of the second memory cell array CA2 and a bit line of the first memory cell array CA1 may be commonly connected to the page buffer of the first peripheral circuit PC1.

[0054] In an embodiment, the fourth contact plug 46B may be electrically connected to a gate line of the second gate structure GST2, and may extend through the second step structure of the second gate structure GST2. The fourth contact plug 46B may pass through the fourth bonding insulating layer 44 and the fourth bonding layer 42 to electrically connect to the fourth bonding pad 45B. The fourth bonding pad 45B may be connected to the second contact plug 36B2 through the second interconnection structure IC2. In this case, the fourth contact plug 46B, the fourth bonding pad 45B, the second contact plug 36B2, and the second bonding pad 35B2 may be electrically connected. Therefore, the gate line 47 of the second memory cell array CA2 may be connected to the row decoder of the second peripheral circuit PC2.

[0055] Referring to FIG. 4B, the semiconductor device may include a first semiconductor structure SS1, a first bonding structure BS1, a second semiconductor structure SS2, a second bonding structure BS2, a third semiconductor structure SS3, and contact plugs 36A, 36B, 36C, 46A, 46B, and 46C.

[0056] A first gate structure GST1 of the second semiconductor structure SS2 may include a first step structure, and a second gate structure GST2 of the third semiconductor structure SS3 may include a second step structure. The first and second step structures may be positioned to overlap. The fourth contact plug 46B may extend through the second step structure, and the second contact

plug **36B** may extend through the first step structure. The fourth contact plug **46B**, the fourth bonding pad **45B**, the second contact plug **36B**, and the second bonding pad **35B** may be commonly connected to the second peripheral circuit **PC2**. Therefore, the first memory cell array **CA1** and the second memory cell array **CA2** may share the second peripheral circuit **PC2**, and the same operation voltage may be applied to the gate lines **37** and **47** positioned on the same layer. [0057] The first bonding structure **BS1** may further include a fifth bonding pad **35C**, and the fifth contact plug **36C** passing through the second semiconductor structure **SS2** may be electrically connected to the fifth bonding pad **35C**. The fifth contact plug **36C** may extend into the first bonding structure **BS1** through the second interlayer insulating layer **IL2**, and may be directly connected to the fifth bonding pad **35C**.

[0058] The second bonding structure **BS2** may further include a sixth bonding pad **45C**, and the sixth contact plug **46C** passing through the third semiconductor structure **SS3** may be electrically connected to the sixth bonding pad **45C**. The sixth contact plug **46C** may extend into the second bonding structure **BS2** through the third interlayer insulating layer **IL3**, and may be directly connected to the sixth bonding pad **45C**. The sixth bonding pad **45C** may be electrically connected to the fifth contact plug **36C** through the second interconnection structure **IC2**.

[0059] The first semiconductor structure **SS1** may further include a third peripheral circuit **PC3**. The third semiconductor structure **SS3** may further include an uppermost metal line **49**. The uppermost metal line **49** may be an external connection pad, a power transmission line, or a signal transmission line. Through the sixth contact plug **46C**, the sixth bonding pad **45C**, the fifth contact plug **36C**, and the fifth bonding pad **35C**, the uppermost metal line **49** and the third peripheral circuit **PC3** may be electrically connected to each other. For example, the third peripheral circuit **PC3** may be a logic circuit, an analog circuit, an input/output circuit, or the like.

[0060] Referring to FIG. 4C, the semiconductor device may include a first semiconductor structure **SS1**, a first bonding structure **BS1**, a second semiconductor structure **SS2**, a second bonding structure **BS2**, a third semiconductor structure **SS3**, a third bonding structure **BS3**, and a fourth semiconductor structure **SS4** in the recited order. The semiconductor device may further include contact plugs **36A**, **36B**, **36C**, **46A**, **46B**, **46C**, **56A**, **56B**, and **56C**.

[0061] The fourth semiconductor structure **SS4** may include a source structure **S**, a third gate structure **GST3**, a channel structure **CH**, a slit structure **SLS**, a fourth interlayer insulating layer **IL4**, and a fourth interconnection structure **IC4**. The third gate structure **GST3** may be positioned on the source structure **S**, and the third gate structure **GST3** may include gate lines **57** and insulating layers **58** that are alternately stacked. In an embodiment, the gate line **57** may be a source select line, a word line, or a drain select line. The third gate structure **GST3** may include a third step structure. The first step structure, the second step structure, and the third step structure may be arranged to overlap.

[0062] The fourth interconnection structure **IC4** may be positioned in the fourth interlayer insulating layer **IL4**. The fourth interconnection structure **IC4** may be connected to the channel structure **CH**, the slit structure **SLS**, the gate lines **57**, and the like, and may include a via and/or a line. In an embodiment, the fourth interconnection structure **IC4** may include a bit line connected to the channel structure **CH**.

[0063] The third bonding structure **BS3** may include a fifth bonding layer **51**, a sixth bonding layer **52**, a fifth bonding insulating layer **53**, a sixth bonding insulating layer **54**, and bonding pads **55A**, **55B**, and **55C**. The contact plugs **56A**, **56B**, and **56C** may extend through the fourth semiconductor structure **SS4**, and may extend into the third bonding structure **BS3** to be directly connected to the bonding pads **55A**, **55B**, and **55C**.

[0064] In an embodiment, the seventh contact plug **56A** may be electrically connected to the bit line, and may extend through the third gate structure **GST3** and the source structure **S**. The seventh contact plug **56A** may pass through the sixth bonding insulating layer **54** and the sixth bonding layer **52** to electrically connect to the seventh bonding pad **55A**. Therefore, a first memory cell

array CA1, a second memory cell array CA2, and a third memory cell array CA3 may share the page buffer of the first peripheral circuit PC1.

[0065] In an embodiment, the eighth contact plug 56B may be connected to the gate line 57 of the third memory cell array CA3, and may extend through the third step structure of the third gate structure GST3. The eighth contact plug 56B may pass through the sixth bonding insulating layer 54 and the sixth bonding layer 52 to electrically connect to the eighth bonding pad 55B. Therefore, the first memory cell array CA1, the second memory cell array CA2, and the third memory cell array CA3 may share the row decoder of the second peripheral circuit PC2.

[0066] In an embodiment, the ninth contact plug 56C may pass through the fourth interlayer insulating layer IL4 to electrically connect to the ninth bonding pad 55C. Therefore, through the ninth contact plug 56C, the ninth bonding pad 55C, the sixth contact plug 46C, the sixth bonding pad 45C, the fifth contact plug 36C, and the fifth bonding pad 35C, an uppermost metal line 59 and the third peripheral circuit PC3 may be electrically connected. For example, the third peripheral circuit PC3 may be a logic circuit, an analog circuit, an input/output circuit, or the like.

[0067] According to the structure described above, three or more semiconductor structures SS1, SS2, SS3, and SS4 may be multi-bonded. The contact plugs 36A, 36B, 36B1, and 36B2 passing through the second semiconductor structure SS2 may extend into the first bonding structure BS1, and may be directly connected to the bonding pads 35A, 35B, 35B1, and 35B2. The contact plugs 46A, 46B, and 46C passing through the third semiconductor structure SS3 may extend into the second bonding structure BS2, and may be directly connected to the bonding pads 45A, 45B, and 45C. The contact plugs 56A, 56B, and 56C passing through the fourth semiconductor structure SS4 may extend into the third bonding structure BS3, and may be directly connected to the bonding pads 55A, 55B, and 55C. Therefore, the bonding structures BS1, BS2, and BS3 may be simplified.

[0068] In addition, the peripheral circuits PC1 and PC2 and the memory cell arrays CA1 and CA2 may be electrically connected through the bonding pads 35A, 35B1, 35B2, 45A, 45B, 55A, and 55B and the contact plugs 36A, 36B1, 36B2, 46A, 46B, 56A, and 56B. Therefore, the first memory cell array CA1 and the second memory cell array CA2 may share the peripheral circuits PC1 and PC2.

[0069] FIGS. 5A to 5D are diagrams illustrating a method of manufacturing a semiconductor device according to an embodiment of the present disclosure. Hereinafter, a description that overlaps the content described above may be omitted.

[0070] Referring to FIG. 5A, a first wafer WF1 including a first bonding layer 61 and a bonding pad 65 may be formed. The first wafer WF1 may further include a first semiconductor structure SS1 such as a peripheral circuit and a memory cell array. The first semiconductor structure SS1 may include a first interconnection structure IC1, and the bonding pad 65 may be connected to the first interconnection structure IC1. The first wafer WF1 may further include a first bonding insulating layer 63 disposed between the first bonding layer 61 and the first semiconductor structure SS1. The bonding pad 65 may pass through the first bonding layer 61 and the first bonding insulating layer 63. The first bonding layer 61 may include SiCN, and the first bonding insulating layer 63 may include an insulating material such as oxide or nitride.

[0071] A second wafer WF2 including a second bonding layer 62 and a sacrificial plug 66 may be formed. The second wafer WF2 may further include a second semiconductor structure SS2 such as a peripheral circuit and a memory cell array. The second wafer WF2 may further include a second bonding insulating layer 64, and the sacrificial plug 66 may pass through the second semiconductor structure SS2, the second bonding layer 62, and the second bonding insulating layer 64. The second bonding layer 62 may include SiCN, and the second bonding insulating layer 64 may include an insulating material such as oxide or nitride. The sacrificial plug 66 may include a material with a high etch selectivity with respect to the second bonding layer 62 and the second bonding insulating layer 64. The sacrificial plug 66 may include a conductive material such as tungsten, polysilicon, titanium nitride, or an insulating material such as nitride.

[0072] Referring to FIG. 5B, the first bonding layer **61** and the second bonding layer **62** may be bonded so that the bonding pad **65** and the sacrificial plug **66** are connected. Through this, a bonding structure BS may be formed. Through bonding, a surface of the first bonding layer **61** and a surface of the second bonding layer **62** may be chemically bonded. In an embodiment, carbon (C) and nitrogen (N) of the SiCN layer may be removed, and Si—O—Si bonding may be generated. The bonding pad **65** and the sacrificial plug **66** may be connected to each other by directly contacting each other without chemical bonding.

[0073] Referring to FIG. 5C, the sacrificial plug **66** may be removed to form an opening OP. The opening OP may pass through the second semiconductor structure SS2, the second bonding insulating layer **64** and the second bonding layer **62** to expose at least partially the top surface of the bonding pad **65**. The opening OP may have a high aspect ratio. Also, the second bonding layer **62** and the second bonding insulating layer **64** may be exposed through the opening OP.

[0074] Referring to FIG. 5D, a contact plug **67** electrically connected to the bonding pad **65** may be formed in the opening OP. The contact plug **67** may be formed by depositing a conductive material along a surface of the bonding pad **65** and an inner surface of the opening OP. A barrier layer **67A** may be formed in the opening OP, and a metal layer **67B** may be formed in the barrier layer **67A**.

[0075] According to the manufacturing method described above, after bonding the first wafer WF1 and the second wafer WF2, the sacrificial plug **66** may be replaced with the contact plug **67**. The bonding pad **65** and the sacrificial plug **66** may not be chemically bonded, and the contact plug **67** may be formed by depositing a conductive material on the bonding pad **65**. Therefore, the bonding pad **65** and the contact plug **67** may be electrically connected even though the bonding pad **65** and the contact plug **67** are not directly bonded.

[0076] In order to directly bond bonding pads, both of the bonding pad **65** and the contact plug **67** are required to be formed of copper, but according to an embodiment of the present disclosure, a material of the bonding pad **65** and the contact plug **67** may not be limited to copper. In an embodiment, the bonding pad **65** and the contact plug **67** may be formed of a material such as tungsten-tungsten of which a use is difficult due to a restriction due to a thermal process, or the bonding pad **65** and the contact plug **67** may be formed of different types of materials such as copper-tungsten.

[0077] FIGS. 6A to 6C are diagrams illustrating a method of manufacturing a semiconductor device according to an embodiment of the present disclosure. FIG. 6A is an enlarged view of a region B of FIG. 5B, FIG. 6B is an enlarged view of a region C of FIG. 5C, and FIG. 6C is an enlarged view of a region D of FIG. 5D. Hereinafter, a description that overlaps the content described above may be omitted.

[0078] Referring to FIG. 6A, the bonding pad **65** and the sacrificial plug **66** may be connected by bonding the first bonding layer **61** and the second bonding layer **62**. At this time, a gap G may exist between the bonding pad **65** and the sacrificial plug **66**. The gap G may be caused by dishing induced in a planarization process for forming the bonding pad **65**. Due to the dishing, an upper surface of the bonding pad **65** may be positioned lower than an upper surface of the first bonding layer **61**. In this case, the first bonding layer **61** and the second bonding layer **62** may be in contact with each other to be bonded to each other, but the gap G may exist between the bonding pad **65** and the sacrificial plug **66**, and the first bonding layer **61** and the second bonding layer **62** may not be in contact with each other.

[0079] Referring to FIG. 6B, the sacrificial plug **66** may be removed to form an opening OP. The opening OP may be connected to the gap G and extend into the first bonding layer **61**.

[0080] Referring to FIG. 6C, the contact plug **67** may be formed in the opening OP. In an embodiment, after forming the barrier layer **67A** in the opening OP, the metal layer **67B** may be formed. In an embodiment, the opening OP may be filled with the metal layer **67B** without forming the barrier layer **67A**.

[0081] According to the manufacturing method described above, the contact plug **67** may be

formed after wafer bonding. Therefore, even though the gap G is formed between the bonding pad **65** and the sacrificial plug **66**, the gap G may be filled with the contact plug **67**. Therefore, the bonding pad **65** and the contact plug **67** may be connected without the gap G, and an ohmic contact may be formed.

[0082] FIGS. **7A** to **7C** are diagrams illustrating a method of manufacturing a semiconductor device according to an embodiment of the present disclosure. FIG. **7A** is an enlarged view of a region B of FIG. **5B**, FIG. **7B** is an enlarged view of a region C of FIG. **5C**, and FIG. **7C** is an enlarged view of a region D of FIG. **5D**. Hereinafter, a description that overlaps the content described above may be omitted.

[0083] Referring to FIG. **7A**, the bonding pad **65** and the sacrificial plug **66** may be connected by bonding the first bonding layer **61** and the second bonding layer **62**. At this time, a foreign substance **71** may exist between the bonding pad **65** and the sacrificial plug **66**. The foreign substance **71** may be a by-product in a manufacturing process, and may be an etching residue, a bonding residue, or the like. The foreign substance **71** may be a polymer-based material, and due to the foreign substance **71**, the bonding pad **65** and the sacrificial plug **66** may not be directly connected or a contact area between them may be reduced.

[0084] Referring to FIG. **7B**, the sacrificial plug **66** may be removed to form an opening OP. The foreign substance **71** may be exposed through the opening OP, and at least a portion of the exposed foreign substance **71** may be removed. In an embodiment, the sacrificial plug **66** may be removed through a deep-out process and then the foreign substance **71** may be removed through an etching process. A portion of the foreign substance **71** exposed by the opening OP may be removed, and a portion interposed between the bonding pad **65** and the second bonding layer **62** may remain without being removed.

[0085] Referring to FIG. **7C**, the contact plug **67** may be formed in the opening OP. In an embodiment, after forming the barrier layer **67A** in the opening OP, the metal layer **67B** may be formed. In an embodiment, the opening OP may be filled with the metal layer **67B** without forming the barrier layer **67A**.

[0086] According to the manufacturing method described above, the contact plug **67** may be formed after wafer bonding. Therefore, even when the foreign substance **71** exists between the bonding pad **65** and the sacrificial plug **66**, the foreign substance **71** may be removed by the process of replacing the sacrificial plug **66** with the contact plug **67**. Therefore, the bonding pad **65** and the contact plug **67** may be connected without the foreign substance **71**, and an ohmic contact may be formed.

[0087] FIGS. **8A** to **8G** are diagrams illustrating a method of manufacturing a semiconductor device according to an embodiment of the present disclosure. Hereinafter, a description that overlaps the content described above may be omitted.

[0088] Referring to FIGS. **8A** and **8B**, a second wafer WF2 including a first sacrificial plug **86** and a second bonding layer **81B** may be formed.

[0089] First, referring to FIG. **8A**, a first preliminary bonding structure **81** may be formed on a second substrate **100**. The first preliminary bonding structure **81** may include a second bonding layer **81B**. The first preliminary bonding structure **81** may further include bonding insulating layers **81A** and **81C**. The second bonding layer **81B** may be interposed between the bonding insulating layers **81A** and **81C**. The bonding insulating layers **81A** and **81C** may be made of an insulating material such as an oxide or nitride suitable for protecting the second bonding layer **81B** in a manufacturing process. The second bonding layer **81B** may form a bonding interface through chemical bonding, and may include SiCN.

[0090] Subsequently, a second semiconductor structure SS2 may be formed on the first preliminary bonding structure **81**. The second semiconductor structure SS2 may include a first memory cell array CA1. The second semiconductor structure SS2 may include a first sacrificial plug **86**, a source structure S, a first gate structure GST1, a channel structure CH, a slit structure SLS, and a

second interlayer insulating layer IL2. The gate structure GST may include gate lines **97** and insulating layers **98** that are alternately stacked. The first sacrificial plug **86** may extend through the first gate structure GST1 and the source structure S, or may extend through the second interlayer insulating layer IL2. The first sacrificial plug **86** may extend into the first preliminary bonding structure **81**. In an embodiment, the first sacrificial plug **86** may pass through the second bonding insulating layer **81C** and the first bonding layer **81B** and contact the top surface of the first bonding insulating layer **81A**.

[0091] In an embodiment, the second semiconductor structure SS2 may include a stack instead of the gate structure GST. The stack may include sacrificial layers instead of the gate lines **97**. The stack may include the sacrificial layers and insulating layers **98** that are alternately stacked. In this case, a sacrificial slit structure may be formed instead of the slit structure SLS.

[0092] Subsequently, a first sacrificial bonding structure **82** may be formed on the second semiconductor structure SS2. The first sacrificial bonding structure **82** may include a first sacrificial bonding layer **82B** and may further include a first sacrificial bonding insulating layer **82A**, with the first sacrificial bonding insulating layer **82A** being disposed between the first sacrificial bonding layer **82B** and the second interlayer insulating layer IL2. The first sacrificial bonding insulating layer **82A** may be formed on the second semiconductor structure SS2.

[0093] Subsequently, referring to FIG. **8B**, a second sacrificial bonding structure **83** may be formed on a carrier wafer **200**. The second sacrificial bonding structure **83** may include a second sacrificial bonding layer **83B**, and may further include a second sacrificial bonding insulating layer **83A**.

[0094] Subsequently, the first sacrificial bonding layer **82B** and the second sacrificial bonding layer **83B** may be bonded and the second substrate **100** and the bonding insulating layer **81A** may be removed. Through this, a second wafer W2 in which the second semiconductor structure SS2 is bonded to the carrier wafer **200** may be formed.

[0095] Referring to FIG. **8C**, a first wafer WF1 including a first bonding pad **85** and a first bonding layer **84B** may be formed. First, the first semiconductor structure SS1 may be formed on a first substrate **300**. The first semiconductor structure SS1 may include a peripheral circuit PC, and the peripheral circuit PC may include a page buffer, a row decoder, a logic circuit, an analog circuit, an input/output circuit, or the like. The first semiconductor structure SS1 may further include a first interlayer insulating layer IL1 and a first interconnection structure IC1.

[0096] Subsequently, a second preliminary bonding structure **84** may be formed on the peripheral circuit PC. The second preliminary bonding structure **84** may include a first bonding layer **84B** and a first bonding pad **85**, and may further include a bonding insulating layer **84a**. The first bonding pad **85** may pass through the first bonding layer **84B** and the bonding insulating layer **84a** to electrically connect to the peripheral circuit PC through the first interconnection structure IC1.

[0097] Subsequently, the first wafer WF1 and the second wafer WF2 may be bonded. The first bonding layer **84B** and the second bonding layer **81B** may be bonded so that the first sacrificial plug **86** and the first bonding pad **85** are connected. Through this, surfaces of the first bonding layer **84B** and the second bonding layer **81B** may be chemically bonded. The first sacrificial plug **86** and the first bonding pad **85** may be in physical contact without chemical bonding. A gap may exist between the first sacrificial plug **86** and the first bonding pad **85**, or a foreign substance may be interposed.

[0098] Referring to FIG. **8D**, the first preliminary bonding structure **81** and the second preliminary bonding structure **84** may be bonded to thus form a first bonding structure BS1. Subsequently, the carrier wafer **200**, the second sacrificial bonding structure **83**, and the first sacrificial bonding structure **82** may be removed to expose the second semiconductor structure SS2 and the first sacrificial plug **86**.

[0099] Subsequently, the first sacrificial plug **86** may be removed to form a first opening OP1 by selective etching. The first opening OP1 is formed to expose the first bonding pad **85**.

[0100] If a gap exists between the first sacrificial plug **86** and the first bonding pad **85**, then the gap

is exposed through the first opening OP1. Likewise, if a foreign substance exists between the first sacrificial plug **86** and the first bonding pad **85**, the foreign substance is also exposed through the first opening OP1.

[0101] Referring to FIG. **8E**, a first contact plug **87** may be formed in the first opening OP1. The first contact plug **87** may be electrically connected to the first bonding pad **85**. When a gap exists between the first sacrificial plug **86** and the first bonding pad **85**, the first contact plug **87** may be formed in the first opening OP1 and the gap. When a foreign substance exists between the first sacrificial plug **86** and the first bonding pad **85**, the first contact plug **87** may be formed after removing the foreign substance. Through this, a bonding defect may be reduced.

[0102] In an embodiment, when the second semiconductor structure SS2 includes a stack and a sacrificial slit structure, the sacrificial slit structure may be removed to form a slit, and sacrificial layers may be replaced with gate lines **97** through the slit. Subsequently, the slit structure SLS may be formed in the slit. Through this, the gate structure GST may be formed.

[0103] Subsequently, a second interconnection structure IC2 connected to the first contact plug **87**, the channel structure CH, the slit structure SLS, and the like may be formed. Subsequently, a third preliminary bonding structure **88** may be formed. The third preliminary bonding structure **88** may include a third bonding layer **88B** and a second bonding pad **89**, and may further include a bonding insulating layer **88A**. The second bonding pad **89** may extend through the third bonding layer **88B** and the bonding insulating layer **88A** to electrically connect to the second interconnection structure IC2. The second bonding pad **89** may be electrically connected to the first contact plug **87**.

[0104] Referring to FIG. **8F**, a third wafer WF3 including a fourth bonding layer **91A** and a second sacrificial plug **92** may be formed. The third wafer WF3 may include a fourth preliminary bonding structure **91**. The fourth preliminary bonding structure **91** may include the fourth bonding layer **91A** and may further include a bonding insulating layer **91B**.

[0105] The third wafer WF3 may include a third semiconductor structure SS3. The third semiconductor structure SS3 may include a second memory cell array CA2. The third semiconductor structure SS3 may include the second sacrificial plug **92**, the source structure S, the second gate structure GST2, the channel structure CH, the slit structure SLS, and the third interlayer insulating layer IL3. The second gate structure GST2 may include gate lines **107** and insulating layers **108** that are alternately stacked. The second sacrificial plug **92** may extend through the second gate structure GST2 and the source structure S, or may extend through the third interlayer insulating layer IL3. The second sacrificial plug **92** may extend into the fourth preliminary bonding structure **91**. In an embodiment, the second sacrificial plug **92** may pass through the bonding insulating layer **91B** and the fourth bonding layer **91A**.

[0106] Subsequently, the third bonding layer **88B** and the fourth bonding layer **91A** may be bonded so that the second sacrificial plug **92** and the second bonding pad **89** are connected. The third bonding layer **88B** and the fourth bonding layer **91A** may be bonded, and thus a second bonding structure BS2 may be formed. When bonding the third wafer WF3, as previously described with reference to FIGS. **8A** to **8C**, a carrier wafer may be used, and the carrier wafer may be removed to expose the second sacrificial plug **92**.

[0107] Referring to FIG. **8G**, the second sacrificial plug **92** may be removed to form a second opening OP2 exposing the second bonding pad **89**. Subsequently, a second contact plug **93** may be formed in the second opening OP2. The second contact plug **93** may be directly connected to the second bonding pad **89**.

[0108] Subsequently, a third interconnection structure IC3 connected to the second contact plug **93**, the channel structure CH, the slit structure SLS, and the like may be formed.

[0109] According to the manufacturing method described above, a sacrificial plug may be formed when forming a wafer, and the sacrificial plug may be replaced with a contact plug after wafer bonding. Therefore, instead of making a bond between bonding pads, the bonding pad and the contact plug may be directly connected. In a conventional method of making a bond between two

bonding pads, the material of the bonding pads is typically limited to copper. However, according to embodiments of the present disclosure, an expanded range of materials may be used for the bonding pad and the contact plug.

[0110] The structure and the manufacturing method according to the above-described embodiments may be applied to semiconductor devices of various structures. FIGS. 9 and 10 illustrate a schematic configuration of a semiconductor device to which the above-described embodiments are applicable.

[0111] FIG. 9 is a configuration diagram of a semiconductor device according to an embodiment of the present invention disclosure.

[0112] Referring to FIG. 9, the semiconductor device may include a substrate SUB, a peripheral circuit PC formed over the substrate SUB, and a memory cell array CA formed over the peripheral circuit PC. In the illustrated embodiment, the peripheral circuit PC and the memory cell array CA may be formed over or on the same substrate SUB.

[0113] The substrate SUB may be made of or include a semiconductor material. In an embodiment, the semiconductor material may include at least one of a group IV semiconductor, a group III-V compound semiconductor, and a group II-VI compound semiconductor. For example, the group IV semiconductor may include single crystal silicon Si, polycrystalline silicon, germanium Ge, or silicon germanium SiGe. The group III-V compound semiconductor may include GaAs, GaN, GaP, GaAsP, GaInAsP, AlAs, AlGa, InP, InSb, or InGaAs. The group II-VI compound semiconductor may include ZnS, ZnO, or CdS.

[0114] The substrate SUB may include a dielectric layer. The substrate SUB may be a silicon-on-insulator (SOI) substrate, a germanium-on-insulator (GeOI) substrate, or a glass substrate. The substrate SUB may include an organic material. In an embodiment, the substrate SUB may include graphene.

[0115] The substrate SUB may be a bulk wafer or an epitaxial layer grown in a selective epitaxial growth (SEG) method. The substrate SUB may be a layer formed in a metal induced lateral crystallization (MILC) method and may partially include metal. The substrate SUB may have a single crystalline, polycrystalline, or amorphous state. The substrate SUB may include an impurity of group II, group III, group IV, group V, or group VI. In an embodiment, the substrate SUB may include an n-well region doped with an n-type impurity and/or a p-well region doped with a p-type impurity.

[0116] The peripheral circuit PC may be disposed between the substrate SUB and the memory cell array CA. The peripheral circuit PC may include a row decoder, a column decoder, a page buffer, a logic circuit, a control circuit, a sense amplifier, an input/output circuit, and the like. In an embodiment, the peripheral circuit PC may include an NMOS transistor, a PMOS transistor, a resistor, a capacitor, and the like. The peripheral circuit PC may further include an interconnection structure. The interconnection structure may be used as a path for transferring an operation voltage, and may include a contact plug, a line, and the like.

[0117] The memory cell array CA may include memory cells. In an embodiment, the memory cell array CA may include memory strings connected between a source line and a bit line, and each memory string may include stacked memory cells. In an embodiment, the memory cell array CA may include memory cells connected between a word line and a bit line. The memory cell array CA may further include an interconnection structure.

[0118] FIG. 10 is a configuration diagram of a semiconductor device according to an embodiment of the present disclosure.

[0119] Referring to FIG. 10, the semiconductor device may include a substrate SUB, a peripheral circuit PC, a bonding structure BS, and a memory cell array CA. For example, the peripheral circuit PC and the memory cell array CA may be respectively formed on separate substrates and then bonded. The semiconductor device may further include a support base SP_B.

[0120] The substrate SUB may be used as a support by the process of forming the peripheral circuit

PC. The support base SP_B may be used as a support by the process of forming the memory cell array CA. In an embodiment, after respectively manufacturing a first wafer including the memory cell array CA and a second wafer including the peripheral circuit PC, the first wafer and the second wafer may be electrically connected by the bonding structure BS. After bonding, at least a portion of the support base SP_B of the first wafer may be removed. The support base SP_B may be completely removed or may partially remain on the memory cell CA array.

[0121] The support base SP_B may be a semiconductor substrate, an insulating substrate, a silicon-on-insulator (SOI) substrate, a germanium-on-insulator (GeOI) substrate, or the like. The support base SP_B may be a bulk wafer, an epitaxial layer grown in a selective epitaxial growth (SEG) method, or a layer formed in a metal induced lateral crystallization (MILC) method. The support base SP_B may have a single crystalline, polycrystalline, or amorphous state. The support base SP_B may include an impurity of group II, group III, group IV, group V, or group VI.

[0122] The bonding structure BS may connect the memory cell array CA and the peripheral circuit PC. In an embodiment, the memory cell array CA and the peripheral circuit PC may be bonded in a wafer-on-wafer bonding method, a chip-on-wafer bonding method, a chip-on-chip bonding method, or the like. The bonding structure BS may include a bonding pad, a bonding layer, a bonding interface, and the like. The bonding pad may include a metal such as copper and aluminum, and/or an alloy. The bonding interface may include a non-metal-non-metal interface, a metal-metal interface, or the like. The memory cell array CA and the peripheral circuit PC may be electrically connected by the bonding structure BS.

[0123] In an embodiment, an interconnection structure included in the memory cell array CA and/or the peripheral circuit PC may be directly connected without a bonding pad. In an embodiment, a bonding layer included in the memory cell array CA and a bonding layer included in the peripheral circuit PC may be bonded to form a bonding interface, and the interconnection structure included in the memory cell array CA and the interconnection structure included in the peripheral circuit PC may be directly connected. Through this, contact plugs, lines, and the like formed on different wafers may be electrically connected without a separate bonding pad.

[0124] Other configurations may be equal or similar to those described above with reference to FIG. 9.

[0125] The semiconductor device may have a structure in which the embodiments described above with reference to FIGS. 9 and 10 are combined or may have a partially modified structure. In the embodiment described with reference to FIGS. 9 and 10, positions of the memory cell array CA and the peripheral circuit PC may be changed. At least one memory cell array CA and/or at least one peripheral circuit PC may be additionally bonded to the embodiment described with reference to FIG. 9. In an embodiment, a portion of the peripheral circuitry PC may be disposed in the memory cell array CA.

[0126] Although embodiments according to the technical concepts of the present disclosure have been described with reference to the accompanying drawings, this is only for describing an embodiment according to the concept of the present disclosure, and the present disclosure is not limited to the above-described embodiments. Within the scope of the technical concepts of the present disclosure, various forms of substitution, modification, change, and combination of the embodiments will be possible by those skilled in the art to which the present disclosure belongs, and these also belong to the scope of the present disclosure. Furthermore, the embodiments may be combined to form additional embodiments.

Claims

1. A semiconductor device comprising: a first semiconductor structure including a peripheral circuit and an interconnection structure electrically connected to the peripheral circuit; a second semiconductor structure; a bonding layer positioned between the first and second semiconductor

structures; a bonding pad electrically connected to the interconnection structure, the bonding pad extending into the bonding layer; and a contact plug extending into the bonding layer through the second semiconductor structure, wherein the contact plug is electrically connected to the bonding pad.

2. The semiconductor device of claim 1, wherein the bonding layer comprises first and second bonding layers, the second bonding layer being positioned between the first bonding layer and the second semiconductor structure, wherein the second bonding layer forms a bonding interface with the first bonding layer, wherein the bonding pad passes through the first bonding layer, and wherein the contact plug passes through the second bonding layer to connect with the bonding pad at the bonding interface.

3. The semiconductor device of claim 1, wherein the second semiconductor structure includes a gate structure, a channel structure extending through the gate structure, and memory cells stacked along the channel structure, and wherein the contact plug extends through the gate structure.

4. The semiconductor device of claim 3, wherein the second semiconductor structure further includes a source structure positioned under the gate structure, and wherein the contact plug passes through the source structure.

5. The semiconductor device of claim 3, wherein the contact plug electrically connects a page buffer of the first semiconductor structure and a bit line of the second semiconductor structure.

6. The semiconductor device of claim 3, wherein the contact plug electrically connects a row decoder of the first semiconductor structure and a gate line of the second semiconductor structure.

7. The semiconductor device of claim 1, wherein the contact plug includes a barrier layer and a metal layer in the barrier layer.

8. The semiconductor device of claim 7, wherein the barrier layer extends between the metal layer and the bonding pad.

9. The semiconductor device of claim 1, wherein the second semiconductor structure includes an interlayer insulating layer and an uppermost metal line, and wherein the contact plug extends through the interlayer insulating layer to electrically connect to the uppermost metal line.

10. The semiconductor device of claim 9, wherein the contact plug electrically connects the peripheral circuit of the first semiconductor structure and the uppermost metal line of the second semiconductor structure.

11. The semiconductor device of claim 1, wherein the contact plug comprises: a through portion passing through the second bonding layer and extending into the second semiconductor structure; and an extension portion expanding into the first bonding layer and having a width greater than that of the through portion.

12. The semiconductor device of claim 1, wherein a lower surface of the contact plug includes a curved surface.

13. The semiconductor device of claim 1, further comprising: a first bonding insulating layer positioned between the first bonding layer and the first semiconductor structure, wherein the bonding pad passes through the first bonding insulating layer.

14. The semiconductor device of claim 1, further comprising: a second bonding insulating layer positioned between the second bonding layer and the second semiconductor structure, wherein the contact plug passes through the second bonding insulating layer.

15. A semiconductor device comprising: a first bonding layer; a first bonding pad passing through the first bonding layer; a second bonding layer forming a bonding interface with the first bonding layer; a first gate structure on the second bonding layer; and a first contact plug extending through the first gate structure and the second bonding layer and electrically connected to the first bonding pad, wherein the first contact plug comprises: a metal layer; and a barrier layer surrounding a sidewall of the metal layer and extending between the metal layer and the first bonding pad.

16. The semiconductor device of claim 15, wherein the barrier layer and the first bonding pad contact.

- 17.** The semiconductor device of claim 15, further comprising: a third bonding layer positioned on the first gate structure; a second bonding pad passing through the third bonding layer and electrically connected to the first contact plug; a fourth bonding layer forming a bonding interface with the third bonding layer; a second gate structure on the fourth bonding layer; and a second contact plug extending through the second gate structure and the fourth bonding layer and electrically connected to the second bonding pad.
- 18.** The semiconductor device of claim 15, further comprising: a page buffer positioned under the first bonding layer and electrically connected to the first contact plug through the first bonding pad.
- 19.** The semiconductor device of claim 15, further comprising: a row decoder positioned under the first bonding layer and electrically connected to the first contact plug through the first bonding pad.
- 20.** A method of manufacturing a semiconductor device, the method comprising: forming a first wafer including a first bonding layer and a first bonding pad; forming a second wafer including a second bonding layer and a first sacrificial plug; bonding the first bonding layer and the second bonding layer so that the first bonding pad and the first sacrificial plug are connected; forming a first opening exposing the first bonding pad by removing the first sacrificial plug; and forming a first contact plug electrically connected to the first bonding pad, in the first opening.
- 21.** The method of claim 20, wherein forming the first wafer comprises: forming a peripheral circuit on the first substrate; forming the first bonding layer on the peripheral circuit; and forming the first bonding pad passing through the first bonding layer and electrically connected to the peripheral circuit.
- 22.** The method of claim 20, wherein forming the second wafer comprises: forming the second bonding layer on a second substrate; forming a gate structure on the second bonding layer; and forming the first sacrificial plug extending through the gate structure and the second bonding layer.
- 23.** The method of claim 22, wherein forming the second wafer comprises: forming a sacrificial bonding layer on the gate structure; bonding a carrier wafer to the sacrificial bonding layer; and removing the second substrate.
- 24.** The method of claim 23, further comprising: after bonding the first bonding layer and the second bonding layer, removing the carrier wafer; and removing the sacrificial bonding layer.
- 25.** The method of claim 20, wherein forming the first contact plug comprises: forming a barrier layer in the first opening; and forming a metal layer in the barrier layer.
- 26.** The method of claim 20, wherein in bonding the first bonding layer and the second bonding layer, the first bonding layer and the second bonding layer contact, and a gap exists between the first bonding pad and the first sacrificial plug.
- 27.** The method of claim 26, wherein forming the first contact plug comprises forming the first contact plug in the first opening and the gap.
- 28.** The method of claim 20, further comprising: forming a third bonding layer on the first contact plug; forming a second bonding pad passing through the third bonding layer and electrically connected to the first contact plug; forming a third wafer including a second sacrificial plug and a fourth bonding layer; bonding the third bonding layer and the fourth bonding layer so that the second sacrificial plug and the second bonding pad are connected; forming a second opening exposing the second bonding pad by removing the second sacrificial plug; and forming a second contact plug in the second opening.
- 29.** The method of claim 20, further comprising: before forming the first contact plug, removing a foreign substance exposed through the first opening.
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